

"MONOLITH": Exploring the Virtual Reality Viewing Experience of Non-linear and Multi-threaded Narrative Structures

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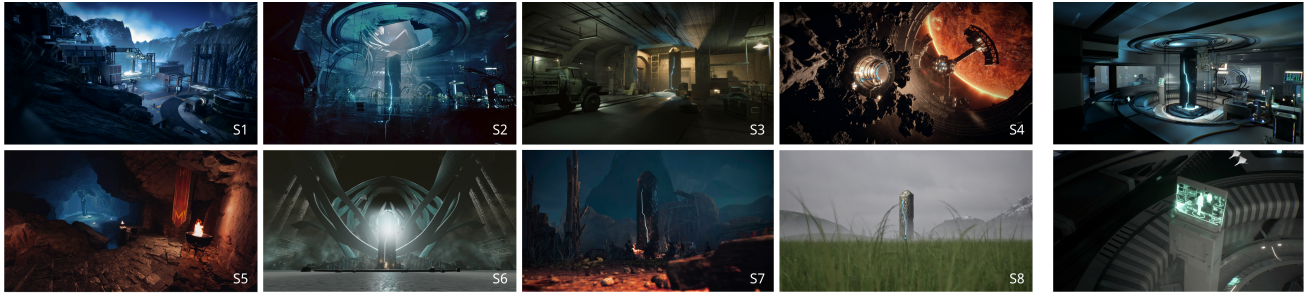
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(a) Overt Narrative Threads from Scene 1 to Scene 8

(b) Covert Threads

Figure 1: (a) Overt narrative threads from Scene 1 to Scene 8. (b) Covert threads.

ABSTRACT

This paper introduces "MONOLITH", a non-linear, multi-threaded narrative film based on VR headset technology. Viewers immerse themselves in the VR experience from a first-person perspective, navigating through time and space to witness stories triggered by a monolith amidst the evolution of civilization. The film uses a multi-threaded structure. The overt thread highlights key events and character interactions with the monolith across different periods, while the covert thread reveals the protagonist's background and identity metaphors. In this science fiction story, viewers explore the complex relationships between time, consciousness, and human nature.

CCS CONCEPTS

• Human-centered computing → Virtual reality; • Applied computing → Arts and humanities.

KEYWORDS

Virtual Reality, Digital Film, Non-linear Narrative, Multi-thread Structure

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1 INTRODUCTION

Non-linear multi-thread narrative is a complex storytelling method that not only departs from traditional linear timelines but also unfolds the story through multiple intertwined narrative threads. In traditional film, non-linear multi-thread narratives attract a wide audience due to their multi-layered innovations in story structure. "MONOLITH" leverages the advantages of VR in scenario construction and non-linear narrative [Liu et al. 2022], aiming to introduce screen-based non-linear multi-thread narratives into the virtual reality experience from a first-person perspective. The film tells the story of Dr. Fang from the MONO organization, who sacrifices his consciousness to a monolith to gain control over time. He becomes trapped in a time loop, observing the indifference and greed behind the facade of human devotion and legacy.

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2 NARRATIVE STRUCTURE

2.1 Non-Linear Narrative

The entire film consists of eight scenes (see Figure1) and has an approximate runtime of ten minutes. It employs a non-linear narrative structure, guiding the audience through a series of stories surrounding the monolith by switching back and forth along the timeline. The film begins with a laboratory scene, which serves as the starting point of the timeline, with subsequent scenes jumping back and forth around this point. The time jumps will progressively span larger intervals and the pacing will accelerate. Each scene represents an independent time period and setting. As viewers traverse these different periods and scenes, they gradually piece together the complete storyline by connecting the information presented.

2.2 Multi-thread Structure

The flashback structure is the most common use of non-linear storytelling in film [Schasché 2010]. In traditional two-dimensional imagery, the interweaving of overt and covert threads is crucial for shaping images or telling stories. In our work, we focus on how to mold a space into overt and covert threads. The film's eight scenes serve as the overt narrative, providing the main story clues and driving the events forward. Flashback segments utilize fixed camera positions, in contrast to the moving shots of the main storyline, to distinguish between the two narrative threads. Whenever the audience transitions to a new period, a brief flashback reveals the protagonist's identity and character relationships. These serve as the covert narrative, deepening the themes and background of the story, enhancing the audience's understanding of the main narrative's motives and emotional depth.

2.3 Information Density

In our film, the unfolding of multiple narrative threads in a non-linear manner necessitates that audiences process a significant amount of information. To address this, we have carefully managed the pace and density of information delivery, adhering to a principle where the middle has the highest density and the beginning and end have lower densities. The middle sections are densely packed with key events and interactions involving the monolith across various historical periods, whereas the opening and closing sections are less dense, primarily serving to introduce and conclude the story, providing necessary transitions and buffering for the audience. The film concludes with a conversation that reveals that the viewer's perspective is actually that of the protagonist's consciousness, leading them to re-examine the film's narrative and connotations.

3 ATTENTION GUIDING DESIGN

The ability of users to autonomously move in a virtual world is a core attribute of traditional VR, whereas in Cinematic VR (CVR), users can only choose viewing angles within the environment and are unable to interact with elements contained within the scene [Mateer 2017]. In order to rationalize the dynamic line of passive movement, we use the jar as a carrier, and the viewer follow the trajectory of the jar throughout the scene. By arranging the spatial elements, we help the viewer to recognize the identity and enhance

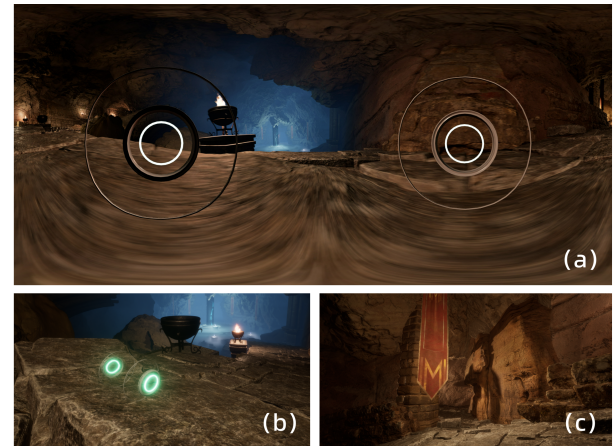


Figure 2: Scene 5. (a) Panoramic rendering. (b) Jar as a carrier. (c) Shadow to guide the viewer's eye.

the immersive experience while maintaining the consistency of the narrative. For example, in Scene 5, we adhere to the physical law of the jar rolling down from a higher ground. However, as the viewing angle shifts, characters engaged in dialogue slowly approach, their swaying shadows cast on the stone wall, and the color palette transitions from warm to cool, hinting at a turning point in the plot. By varying colors, motion, lighting, and sound, we attract viewers' attention and guide them towards the direction in which the story conveys information (see Figure2).

4 CONCLUSION

In this paper, we explore the non-linear multi-thread narrative structure of the VR film "MONOLITH". By employing unique narrative techniques and guided design, the film enhances viewers' immersion and understanding of the story, demonstrating the potential of virtual reality in non-linear multi-thread narratives. This exploration not only enriches the expressive methods of VR narratives but also provides new developmental insights for the future of non-linear narratives in film.

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